

Answer Key — Physics Grade 10

Sub-Strand 2.1: Properties of Waves • Lesson 1: Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
C	C	B	A	B	D	D	A	C

USE AFTER: PREDICT PHASE — Step 2: Live demonstration and Individual PREDICT card ('What did you NOTICE?')

Q1. A matatu travelling in the opposite direction passes your bus on the Nakuru–Nairobi highway, and the driver does not change the horn at all. Which statement is an OBSERVATION (something you actually notice) rather than a guess about the cause?

- A. The horn changes its note because the matatu is tired after a long journey.
- B. The driver pressed the horn harder as the matatu came closer.
- C. The horn sounded higher-pitched as the matatu came towards you and lower-pitched as it moved away.**
- D. The horn button on the matatu was faulty.

Correct answer: C. Noticing what was heard (higher pitch approaching, lower pitch going away) is an observation. The other choices are guesses about the surface features of the matatu or driver, and the driver did not change the horn.

USE AFTER: PREDICT PHASE — Individual PREDICT card: separating 'NOTICE', 'PUZZLING' and 'BEST GUESS'

Q2. On the PREDICT card, the prompt 'What did you NOTICE?' asks for a fact about the bus journey. Which entry is a NOTICE and not a guess about the cause?

- A. The radio receiver was faulty.
- B. Classic 105 has a weaker transmitter than the AM station.
- C. On entering the Kabete underpass, the FM signal (Classic 105, 105.2 MHz) cut out almost completely while the AM signal stayed audible.**
- D. AM waves are stronger than FM waves.

Correct answer: C. Only this choice describes what happened on the journey. The other choices are untested guesses about causes, and a faulty receiver does not fit, because AM still worked on the same receiver.

USE AFTER: OBSERVE PHASE — Station A: PhET Sound Simulation observation table

Q3. At Station A you use the PhET Sound simulation, first moving a listener past a stationary source and then moving the source past a stationary listener. What should you write in your observation table for each set-up?

- A. The speed of a matatu on the highway
- B. The perceived pitch (higher, lower or same) for that set-up**
- C. The reason why the pitch changes
- D. The name of the scientific effect that causes the change

Correct answer: B. Station A is for observing and recording only. The table asks for the perceived pitch (higher, lower or same) for each configuration, and explanations come later.

USE AFTER: OBSERVE PHASE — Station B: Diagram Analysis Card (tunnel width and FM/AM wavelength scales)

Q4. The Station B card shows the Kabete tunnel drawn to scale. The tunnel is about 8 m wide, the FM (105.2 MHz) wavelength is about 2.9 m and the AM wavelength is about 300 m. Which statement correctly reads these numbers?

- A. The AM wavelength is about 300 m, which is much longer than the tunnel width of about 8 m, while the FM wavelength of about 2.9 m is shorter than the tunnel width.**
- B. Both the FM and the AM wavelengths are shorter than the tunnel width.
- C. The AM and FM wavelengths are about the same size as each other.
- D. The FM wavelength is longer than the tunnel width, and the AM wavelength is shorter than the tunnel width.

Correct answer: A. Comparing the numbers: 300 m is greater than 8 m, and 2.9 m is less than 8 m. This only reads the data and does not explain the blackout.

USE AFTER: OBSERVE PHASE — Whole-class debrief: 'WHAT WE SAW' sticky-note board and EVIDENCE column

Q5. During the 'WHAT WE SAW' debrief, which sticky-note entry is a specific, measurable OBSERVATION that could go in the EVIDENCE column?

- A. The matatu horn is broken.
- B. FM wavelength is about 3 m and the tunnel is about 8 m wide.**
- C. Classic 105 is a bad radio station.
- D. FM is blocked because the tunnel is too small.

Correct answer: B. An observation states measurable facts from the evidence, such as the wavelength and the tunnel width. The other choices are explanations or opinions, which are not written under EVIDENCE.

USE AFTER: EXPLAIN PHASE — Gallery walk: dot-sticker voting on 'Best Current Explanation' sentences

Q6. During the gallery walk, each student places one dot sticker on the group explanation they find most convincing. Which explanation should get your dot?

- A. The one that blames a technology fault, such as 'Classic 105 has a weaker transmitter', with no evidence
- B. The one written by your best friend
- C. The one that is the longest
- D. The one that uses evidence from the stations, such as the direction of movement or a measurement**

Correct answer: D. Students are asked to choose the most evidence-based explanation. Length, friendship and claims with no evidence do not make an explanation convincing.

USE AFTER: DQB PHASE — Driving Question Board: student sticky-note questions under 'The Horn Puzzle' and 'The Tunnel Puzzle'

Q7. When you write a question for the Driving Question Board, which question best fits the instruction 'start with HOW or WHY, about the phenomenon, and something you cannot yet answer'?

- A. Where is the Kabete underpass on the map?
- B. Is Classic 105 a good radio station?
- C. Which Sacco owns the matatu?
- D. Why does the FM signal cut out in the tunnel while the AM signal does not?**

Correct answer: D. This question starts with 'Why', is about the phenomenon and can be investigated. The other choices are opinions or simple facts that do not drive an investigation.

USE AFTER: MODEL BUILDING PHASE — Initial Wave Model drawing and pair comparison

Q8. You are asked to draw an 'Initial Wave Model' of a wave travelling from a matatu horn to your ear and entering a tunnel. Which statement about this model is correct?

- A. It only needs to show your current thinking, with labels and arrows, because you will revise it in later lessons.**
- B. The teacher will mark it wrong if the wave shape is not right.
- C. It should copy the teacher's example exactly.
- D. It must be completely correct before you hand it in.

Correct answer: A. The lesson tells students the model does not need to be correct, only to show their current thinking, and that it will be revised. This keeps their initial ideas visible for later comparison.

USE AFTER: END OF LESSON — Closing link of the Model phase: 'The driving question is still open'

Q9. At the end of this first lesson, your class has noticed patterns and posted questions about the matatu horn and the Kabete tunnel. What is the status of the driving question 'Why does the horn change pitch, and why does FM fade in the tunnel?'

- A. It is answered because the PhET simulation showed the pitch change.
- B. It is fully answered because the class voted on the best explanation.
- C. It is still open: the class has observations and first ideas, and will gather more evidence in later lessons.**
- D. It is answered because the diagram shows the FM wavelength and the tunnel width.

Correct answer: C. This is the launch lesson. Students have gathered observations and first ideas only, and the teacher says 'The driving question is still open'. The full explanation comes after more investigation.